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Influence of altitude on local adaptation in upland tree species from central Argentina

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Summary

Steep climatic gradients boost morphological and physiological adjustments in plants, with consequences on performance. The three principal woody species of the Sierras Grandes Mountains of central Argentina have marked differences in sapling performance along their altitudinal distribution. We hypothesize that the steep gradient of climatic conditions across the species' altitudinal distribution promotes

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trait differences between populations of different altitudes that are inherited by the following generation.

Seeds from different altitudes were exposed to three temperature regimes to assess differential germination responses. Saplings were then transplanted to a greenhouse to assess possible variations in attributes and performance after 18 months.

The three species showed differences in germination responses to temperature among altitudes and/or in sapling attributes and performance. In *Maytenus boaria* and *Escallonia cordobensis*, germination success was higher under high temperatures for the highest-altitude, whereas lower temperatures boosted germination of the lowest altitudes. *Polylepis australis* showed no differences in germination among temperature treatments. In the greenhouse, saplings of the three species from intermediate altitudes showed high performance, whereas the upper and lower populations seemed to be adjusted to tolerating more stressful conditions (i.e., lower temperatures at the upper end and water stress at the lower end), showing lower performance toward both altitudinal limits. These patterns agree with those described for saplings growing under field conditions, suggesting adjustments in response to environmental changes undergone by populations along the altitudinal range. The marked adjustments of populations to the local environment suggest a potentially high impact of climatic change on species distribution.

Running head: Local adaptation in mountain trees

Keywords: altitudinal gradient, intraspecific variation, functional traits, germination success, *Polylepis australis*, *Maytenus boaria*, *Escallonia cordobensis*

Introduction

Mountain environments are characterized by marked climatic gradients defined by changes in temperature, atmospheric pressure, UV radiation, precipitation, and length of growing season (Körner 1999). Such climatic factors determine gradients of environmental stress that are important drivers of morphological and physiological variability in plants (Schöb *et al.* 2013; Frei *et al.* 2014; Grassein *et al.* 2014). For instance, leaves of mountain plants generally decrease in length, width and area with increasing altitude, whereas leaf nutrient content per area and leaf thickness increase (Bresson *et al.* 2011; Körner 2012; Vitassé *et al.* 2013). Regenerative traits also vary among populations in response to altitudinal gradients (reviewed by Baskin & Baskin 1998), with selection being expected to favor the local synchronization of germination with optimal climate conditions for seedling survival (Bischoff *et al.* 2006; Galloway & Etterson 2007; Donohue *et al.* 2010; Bischoff & Müller-Schärer 2010; Giménez-Benavides & Milla 2013; García-Fernández *et al.* 2013).

This tight relationship between gradients of environmental stress and plant strategies and distribution that characterize mountains make this system highly vulnerable to environmental changes associated with global warming (Beniston 2003; Gottfried *et al.* 2012; Elsen & Tingley 2015). Locally differentiated shifts in climate might result in populations either going locally extinct, persisting by spatially migrating across ecological niches, or adapting to new conditions in current locations

(Aitken *et al.* 2008; Nicotra *et al.* 2010). Many species are expected to be unable to track climate change, since the currently observed migration rates of many species are several orders of magnitude lower than the anticipated isotherm shift under the future climate change scenario (Davis & Shaw 2001; Giam *et al.* 2010; Pimm *et al.* 2014). Consequently, a decline of cold-adapted species has been predicted in mountain systems under increasing temperatures (Gottfried 2012) especially in mountains with restricted areas (Elsen & Tingley 2015).

Local trait variation can predict a plant species' potential to adapt to variable environmental conditions within a given area (Byars, Papst & Hoffmann 2007; Nicotra *et al.* 2010). Plants with broad altitudinal distribution ranges are exposed to different environmental conditions across the gradient; this situation may lead to variation among populations, ultimately producing local adaptations (Körner 2012; Valladares *et al.* 2014). While irreversible physiological and morphological adjustments or local adaptation might benefit species in the maternal environment, they might also reduce the capability to respond to climatic shifts (Thuiller *et al.* 2008; Nicotra *et al.* 2010; Körner 2012). Accordingly, it is necessary to understand intra-specific variation between different altitudinal populations to improve predictions on the responses of mountain species to changing climatic conditions (Thuiller *et al.* 2008; Körner 2012).

Woodlands of the upper vegetation belt of the Sierras Grandes Mountains of Córdoba (central Argentina) are dominated by *Polylepis australis*, with a lower abundance of *Maytenus boaria* and *Escallonia cordobensis*. These woodlands are particularly restricted in terms of altitudinal retreat, because mountains do not extend

above the alpine zone and there is intensive browsing by livestock affecting sapling performance of the three species (Marcora *et al.* 2013). Previous studies revealed that several traits and components of performance of the three species change along their altitudinal distribution (Marcora *et al.* 2008; 2013). Whether such differences are inherited to the following generations, contributing to population performance and persistence, remains to be tested. Any evidence may provide clues about the ability of species to persist in particular areas after the occurrence of climatic changes (Herman & Sultan 2011), persistence ability has been insufficiently explored in species of subtropical latitudes (Körner 2012).

Here we assessed whether the three principal woody species from this mountain system show differences among populations from different altitudes, which would be linked to their success in the early stages of their life cycles (i.e. seed germination as well as sapling attributes and performance). We hypothesized that the steep gradient of climatic conditions across the altitudinal distribution of the three species promotes trait differences between populations of different altitudes that are inherited by the following generation. We performed an experimental approach to test this hypothesis, which consisted in exposing seeds from different altitudes to three temperature regimes (temperatures from home and two other altitudinal belts) with the aim of assessing possible differential germination responses (e.g. high germination rate at the home temperature). Thereafter, saplings were transplanted to a greenhouse to assess variations in their attributes or performance among altitudes.

Material and Methods

STUDY AREA AND SPECIES

Seeds were collected from the Sierras Grandes de Córdoba, central Argentina. Seeder trees were selected along an altitudinal gradient ranging from 900 m a.s.l. to 2700 m a.s.l., the highest altitude of the mountain range (Linderos road, 32°5'S, 64°9'W). Along the gradient, several climatic variables are highly correlated with altitude, such as mean annual temperature, which decreases from 15.7°C at 900 m a.s.l. to 7.4°C at 2700 m a.s.l. ($r^2 = -0.99$; Marcora *et al.* 2008) and annual precipitation, which increases from 600 to 1000 mm between 600 and 2200 m a.s.l. ($r^2 = 0.67$; Colladon 2014). These changing conditions lead to consistent environmental shifts along the gradient, with a significant decrease in soil temperature ($r^2 = -0.97$) and increasing soil moisture ($r^2 = 0.95$), towards the upper portions of the gradient (Pais-Bosch *et al.* 2012; Tecco *et al.* 2016), suggesting that climatic stress along the altitudinal gradient results from a combination of opposing gradients of temperature and moisture (Tecco *et al.* 2016).

Polylepis australis BITTER (Rosaceae) is a tree endemic to the mountains of central and northwestern Argentina (Kessler & Schmidt-Lebuhn 2006). It grows up to 16 m in height (Renison *et al.* 2013); the species is self-compatible (Seltmann *et al.* 2009), with its pollen and one-seeded nutlets (hereafter referred to as seeds) being wind-dispersed. *Maytenus boaria* MOLINA (Celastraceae) is distributed across the temperate regions of Chile and Argentina, including the central Argentine mountains, as well as Peru and Brazil (Donoso & Wendler 1985). This tree species reaches between 5 and 8 m in height, and produces insect-pollinated flowers with abundant pollen and bird-dispersed fruits (one- to two-seeded capsules) (Cabello & Camelio

1985). The scrub *Escallonia cordobensis* (KUNTZE) HOSSEUS (Escalloniaceae) is endemic to the Sierras de Córdoba and reaches 4 m in height. To the best of our knowledge, pollen or seed dispersal of this species has not been studied, but the white flower type suggests insect pollination.

SEED COLLECTION

Mature fruits of the three species were collected from populations at different altitudes between February and April 2012. The altitudinal range covered for each species was constrained by the availability of accessible adult trees. Fruits of *P. australis* were collected from seven altitudes (900, 1200, 1500, 1800, 2100, 2400 and 2700 m a.s.l.), and those of *E. cordobensis* and *M. boaria* were sampled at four altitudes (900, 1200, 1500 and 1800 m a.s.l.). At each altitude, fruits were collected from a minimum of 20 seeder trees separated from each other by at least 100 m. Seeds of each species and altitude were pooled. Although this methodology does not allow us to evaluate differences among individuals from the same altitude, seed sampling from several, widely-spaced parents allows for the inclusion of a much more representative sample of genotypes from each population and hence would increase ecological realism of the results (Gianoli & Valladares 2012). Fruits of *P. australis* and *E. cordobensis* were stored in paper bags under room temperature conditions; those of *M. boaria* were stored in wet sand at 4°C until the start of germination experiments (December 2012).

GERMINATION TESTS AND SAPLING TRAITS

For the germination experiments, 40 seeds of *P. australis* and *M. boaria* and 50 seeds of *E. cordobensis* were put on filter paper in standard Petri dishes of 9 cm in diameter and kept permanently moist with de-ionized water. Filter paper was replaced weekly to reduce infection by fungi. In the case of *M. boaria*, fruit cover delays germination (Cabello & Camelio 1985); hence, seeds were previously scarified with sulfuric acid at 98% (HSO_4 [98%]) for 15 minutes.

Germination experiments were conducted in germination chambers under an alternating photoperiod of 12h warm white light and 12h darkness. Three temperature treatments were programmed to simulate natural site conditions from the highest to the lowest altitudinal belts of Sierras Grandes, as follows: 8/4 °C, 20/10 °C, and 32/20 °C. These three thermoperiods mimic the day-night temperatures in spring time at the higher, intermediate and lowest ends of the gradient, respectively (Marcora *et al.* 2008; Tecco *et al.* 2016). This experiment follows the reciprocal transplants approach in which populations are exposed to the temperature of the altitudinal belt of origin and temperatures from other altitudinal belts (Kawecki & Ebert 2004). For each temperature treatment, six Petri dishes were used per altitude per species (40×6×3=720 seeds for *P. australis* and *M. boaria*; 50×6×3=900 seeds for *E. cordobensis*). Germinated seeds (with at least 2 mm in radicle length) were removed every two to three days and kept in Petri dishes at 20/10 °C for the greenhouse experiments. Germination experiments were ceased after 60 days, following regular protocols (Baskin & Baskin 1998). At that moment, germination under the cold temperature regime had not plateaued for *E. cordobensis* and *M. boaria*.

To characterize populations, 50 seeds per altitude per species were subjected to a TTC test (Triphenyltetrazolium Chloride [1%]) to assess initial seed viability. Additionally, seed mass of *P. australis* and *M. boaria* was calculated by weighing 30 seeds per species per altitude. Seed mass of *E. cordobensis* was measured in three groups of 200 seeds per altitude because seeds are very small and light (<0.05 mg).

Between three and four weeks after germination, seedlings were transferred to plastic pots containing a compost-sand mixture (2:3) and cultivated at 22 °C and 60 % humidity. Sapling traits and performance measurements (plant height, leaf chlorophyll concentration, leaf area and specific leaf area) were recorded on three dates (6, 12 and 18 months after planting). After sapling harvest (at 10 cm from ground level), leaf and plant aerial biomass was measured. Specifically, growth height (cm) was measured on the longest sapling sprouts. Total leaf chlorophyll concentration was estimated using the SPAD502 chlorophyll meter; SPAD measurements were converted into chlorophyll concentrations ($\mu\text{g}/\text{cm}^2$) using the general relationship provided by Coste *et al.* (2010). This trait was determined as an average of SPAD values from three leaves per individual. Leaf area (LA ; cm^2) and specific leaf area (SLA ; cm^2/g) were determined as the average of 10 whole leaves from the middle region of the sapling, following standard protocols (Pérez-Harguindeguy *et al.* 2013), and their area was calculated using the Win FOLIA Pro S program (Regent Instruments Inc.). Sapling mortality of the three species was recorded between 6 and 18 months after planting in the greenhouse. The number of plants of each altitude and species used for trait measurements are indicated in the figures (see Results).

DATA ANALYSIS

Differences in seed mass between provenances were tested using univariate General Linear Models. Differences in germination success (%) and seed viability between altitudes and temperature treatments were analyzed using Generalized Linear Models (GLMs) with quasi-binomial error structure of data and *Logit* as link function. Altitude and temperature treatments were included as fixed factors, and interaction between factors was included to evaluate if altitudes show differential response to temperatures, i.e. if seeds from higher populations show higher germination success at a different temperature than seeds from lower altitudes. Whenever appropriate, a DGC's *a posteriori* test (Di Rienzo *et al.* 2016) was applied to evaluate significant differences between altitude and temperature treatments.

Variations in sapling mortality among altitudes were analyzed with GLMs using a binary distribution and *Logit* as link function. We used alive and dead saplings of each population as independent samples, so the response variable was binary (alive/dead) and altitude was included as fixed factor.

Differences in sapling traits (height, LA, SLA, chlorophyll concentration, and sapling and leaf dry biomass) among altitudes were analyzed using General Linear Models, with altitude as fixed factor. Some response variables were $\log_{10}$ or square-root transformed to obtain a normal distribution of residuals. In addition, we assessed the relationship of each sapling trait with altitude using Linear Regression and Binary Logistic Regression for mortality. After visual inspection of the graphs, we added the term “altitude squared” where necessary to determine whether variables had an optimum at intermediate altitudes. All statistical analyses were performed

using Infostat (Di Rienzo *et al.* 2016) with its platform to R, and graphs were made in R 3.2.4 (R development team 2016).

Results

EFFECTS OF ALTITUDE ON GERMINATION RESPONSE TO TEMPERATURE

Germination success of the three species differed among altitudes, with some of them showing different germination responses to temperature treatments (Table 1). For *P. australis*, germination success increased with increasing altitudes (Fig. 1a), whereas for *M. boaria* and *E. cordobensis*, the highest germination percentages were recorded at an intermediate altitude (1500 m a.s.l.; Fig 1b and c).

Germination success of *P. australis*, showed differences among altitudes, without statistically significant differences between temperature treatments or their interaction (Fig. 1a; Deviance_A=227.54, DF=6, $P_A < 0.0001$; Deviance_T=8.14, DF=2, $P_T = 0.056$; Deviance_{A*T}=23.12, DF=12, $P_{A*T} = 0.17$). When germination success was analyzed including interaction between altitude and temperature treatments alone, without including the principal effects of both factors, the population from 2100 m a.s.l. showed significant differences in temperature treatments, i.e. germination success was lower under low temperatures than under high temperatures.

In *M. boaria*, there were significant differences in germination success between temperature treatments for all altitudes (Fig. 1b; Deviance_A =46.47, DF=3, $P < 0.0001$; Deviance_T=124.08, DF=2, $P_T < 0.0001$; Deviance_{A*T} =58.35, DF=6, $P_{A*T} < 0.0001$). Seeds from the lowest to intermediate altitudes (i.e. from 900 to 1500 m a.s.l.) showed higher germination success under intermediate temperatures

(20/10°C), whereas seeds from the highest altitude (1800 m a.s.l.) had higher germination success under warm conditions (32/20°C; Fig. 1b). Under cold conditions, seeds from all four altitudes showed lower germination success than at higher temperatures.

In *E. cordobensis*, significant differences in germination success were observed between temperature treatments for all altitudes (Fig. 1c; Deviance_A=167.19, DF=3, P_A<0.0001; Deviance_T=1319.09, DF=2, P_T<0.0001; Deviance_{A*T}=240.54, DF=6, P_{A*T}<0.0001). Populations from all altitudes showed the lowest germination success under the coldest temperature regime. Seeds from the highest altitudes (1800 m a.s.l.) showed higher germination success under warm (32/20 °C) than under intermediate temperatures, whereas the opposite pattern was observed for the lowest populations (900 m a.s.l.; Fig. 1c). Seeds from the two intermediate altitudes (1200 and 1500 m a.s.l.) did not differ in germination success between intermediate and warm temperature treatments (Fig.1c).

Initial seed viability (assessed via TTC test) of *P. australis* and *M. boaria* was different among altitudes, whereas there were no differences for *E. cordobensis*. Seed viability was low in *P. australis* for populations from 900, 1200, 1500 and 2400 m a.s.l. (Deviance=23.99; DF=6; P<0.001), whereas in *M. boaria*, the lowest seed viability was observed for the population from 1500 m a.s.l. (Deviance=10.12; DF=3; P=0.018). Seed mass of the three species varied along the altitudinal gradient (Table 1). *P. australis* exhibited a hump-shaped pattern in seed mass along the gradient (Fig. 2 a), and seed mass of *M. boaria* and *E. cordobensis* decreased with increasing altitude (Fig. 3a and 4a).

EFFECTS OF ALTITUDE ON SAPLING TRAITS AND PERFORMANCE

We found significant differences in several sapling traits and performance components among altitudes for the three species grown in the greenhouse. These differences were consistent on the three sampling dates (i.e. 6, 12 and 18-month-old saplings). Thus, for simplicity, we only report results of the 18-month-old saplings.

For *P. australis*, there were significant differences among altitudes in mortality, sapling dry biomass, height and chlorophyll concentration, as well as in leaf dry biomass, LA and SLA. In most cases, trait variations showed unimodal responses to altitude (Table 1, Fig. 2). The offspring from the lowest and highest sites of the altitudinal gradient had the highest mortality and showed attributes indicative of the overall lowest growth performance (lowest sapling biomass, height and LA), despite their higher SLA values (Fig. 2). In addition, significantly lower chlorophyll concentrations were found in saplings from lower altitudes (900 and 1200 m a.s.l.) than in those from higher altitudes.

In *M. boaria*, sapling dry biomass, height, leaf biomass and LA varied significantly among altitudes (Table 1). Attributes related to increased growth performance (i.e. high values for sapling height and biomass, large leaves) were mostly observed in saplings from lower or intermediate altitudes, whereas the opposite was observed for saplings from higher altitudes (Fig. 3).

Saplings of *E. cordobensis* did not differ significantly in traits among altitudes (Table 1). However, sapling dry biomass showed a unimodal pattern, with the highest biomass occurring in saplings from 1500 m a.s.l. and the lowest, in saplings from 1800 m a.s.l. (Fig. 4).

Discussion

Our experimental approach shows that, the main three woody species from the upper vegetation belt of the Sierras Grandes in central Argentina exhibit differences in germination as well as in sapling attributes and performance among elevational provenances when grown under standard conditions in the greenhouse. These differences were in line with our hypothesis and suggest adjustments of populations in response to environmental changes occurring along an altitudinal gradient (Körner 2012; Grassein *et al.* 2014). Such adjustments might be adaptive when the offspring environment is predictable and similar to that of the maternal environment (Galloway & Etterson 2007; Bischoff & Müller-Schärer 2010; Donohue *et al.* 2010); however, those adjustments might hinder rapid population response to climatic shifts (see below). In the Sierras Grandes, low temperatures and a short growing season would be the main constraints to seedling establishment at high altitudes, whereas decreasing soil moisture, increasing temperatures and water stress would limit sapling establishment toward lower altitudes (Pais-Bosch *et al.* 2012; Marcora *et al.* 2013; Tecco *et al.* 2016).

DIFFERENCES IN GERMINATION SUCCESS IN RESPONSE TO TEMPERATURE AMONG SEEDS FROM DIFFERENT ALTITUDES

For *Maytenus boaria* and *Escallonia cordobensis*, the highest germination success occurred under different temperatures among altitudes. For *Polylepis australis*, only the populations at 2100 m a.s.l. showed differences in germination response to temperature. Exposure of mother plants to certain environmental factors, such as temperature and photoperiod, have shown to promote changes in their seeds, and those determinants (either genetic or non-genetic) in turn affect

germination response and seedling establishment (reviewed in Baskin & Baskin 1998; Giménez-Benavides, Escudero & Iriando 2007; Bischoff *et al.* 2006). Accordingly, the observed differences in germination response to temperature among populations of these species might show adaptation to local environmental conditions (Dunbabin & Cocks 1999; Galloway 2005; Donohue *et al.* 2010) that would favor recruitment under analogous conditions of mother plants (see below).

Our results for *M. boaria* and *E. cordobensis* show that germination success in populations from the upper extreme of the species distribution tends to improve with increasing temperatures, whereas populations from lower altitudes seem to be boosted by cooler temperatures. Although these results may first seem counterintuitive, this would be suggesting an adaptative response to local environmental conditions for successful seedling establishment, by avoiding the foremost constraint that characterizes each extreme of the altitudinal gradient (i.e., germination stimulated by the most benign conditions). On the one hand, the patterns observed in upper populations are similar to those found for other species in cold environments; indeed, some populations avoid germination during or immediately after severe winters (e.g. Giménez-Benavides & Milla 2013), and germinate only under relatively high temperatures. For example, species from northern Europe need higher relative temperatures for germination to occur than Mediterranean species (Fenner & Thompson 2005), which is in turn consistent with findings reported for Arctic species (Baskin & Baskin 1998). On the other hand, towards the warmer extreme of the species distribution range, drought is expected to be the main hazard for seedlings (Fenner & Thompson 2005; Engelbrecht *et al.* 2007; Giménez-Benavides, Escudero & Iriando 2007) and reduced germination in

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response to high temperatures may preclude unsuccessful seedling establishment in populations during unfavorably dry conditions (e.g., Levine, McEachern & Cowan 2008), thereby ensuring local persistence of peripheral populations (Giménez-Benavides, Escudero & Iriondo 2007).

Contrarily, populations from different altitudes of *P. australis* showed no differences in germination among temperature treatments, suggesting no local adaptation in germination response to temperature. Consequently, germination success of *P. australis* would be less affected by changes in temperature than that of the other two species. For *M. boaria* and *E. cordobensis*, germination success of populations might be favored only under maternal environment, but might not be able to endure rapid climatic shifts. As mentioned above, the adaptive significance of such responses can be related to particular habitat requirements along the altitudinal distribution (Bradshaw 1965).

Germination success among populations of *P. australis* and *E. cordobensis* reached values similar to those of their initial viability, but was lower for *M. boaria*. Possible changes in the requirements of cold stratification for breaking dormancy among populations should be assessed in the latter species (Giménez-Benavides & Milla 2013). Finally, germination response of the three species does not seem to be affected by seed mass, as suggested by the different altitudinal patterns of each variable.

DIFFERENCES IN SAPLING ATTRIBUTES AND PERFORMANCE AMONG OFFSPRING OF DIFFERENT ALTITUDES

Populations from different altitudes grown in the greenhouse showed differences in sapling attributes and performance components. Such differences would be due to irreversible physiological and morphological adjustments or to local adaptation (Herman & Sultan 2011; Pigliucci, Murren & Schlichting 2006; Körner 2012). In any case, the differences in sapling traits and performance components among altitudinal populations observed in this study would be related to environmental conditions endured by mother plants, as suggested by the similar responses of plants under greenhouse conditions to those of their counterparts cultivated in the field across an altitudinal gradient (Marcora *et al.* 2013).

Most of *P. australis* sapling traits that showed differences among altitudes followed a unimodal pattern, with attributes indicative of increased performance in saplings from intermediate altitudes (i.e. the largest saplings with the highest chlorophyll content and the lowest mortality). Similarly, the highest sapling height of *M. boaria* and the highest sapling biomass of *E. cordobensis* corresponded to populations from an intermediate altitude (1500 m a.s.l.), also describing a unimodal patterns in response to elevation. Accordingly, a unimodal response to altitude was found for different components of performance of adult trees of *P. australis* growing along the altitudinal gradient (Marcora *et al.* 2008) and for saplings of *P. australis* and *E. cordobensis* grown under field conditions (Marcora *et al.* 2013).

Furthermore, saplings of the three species from both extremes of the species distribution seem to be adjusted to tolerate the more stressful conditions that prevail at both ends of the gradient due to the opposing gradients of temperature and moisture (i.e. lower temperatures at the upper end and water stress at the lower end; Marcora *et al.* 2013; Tecco *et al.* 2016). This adjustment would have negative consequences on performance due to trade-offs in strategies for resource acquisition and use (Wright *et al.* 2005; Reich 2014; Stahl, Reu & Wirth, 2014). Accordingly, traits from both extremes of the altitudinal gradient appear to be governed by security rather than vigour (Körner 2012). Reduced sapling growth observed in *M. boaria* and *E. cordobensis* and reduced leaf size and biomass observed in *P. australis* from high altitudes might be induced either by a shorter length of growing season or by the direct influence of low temperatures and high radiation stress endured by seeder trees (Cornelissen *et al.* 2003; Körner 2012; Stahl, Reu & Wirth, 2014). These intra-specific variations in response to altitude have been assumed to be an adaptation, consisting of enhanced photosynthesis at lower temperatures and increased resistance to freezing temperatures (Goldstein, Rada & Azócar 1985). Likewise, the decreased performance of populations from lower altitudes would be related to water stress, which may be considered a strong selective pressure at the lower limit of the species distribution (Cornelissen *et al.* 2003; Engelbrecht *et al.* 2007; Jump, Mátyás & Peñuelas 2009; Normand *et al.* 2009; Alberto *et al.* 2013). A similar influence of water stress on tree performance at the lower limit was found for other mountain species, such as *Polylepis sericea* (Colmenares-Arteaga, Rada & Luque 2005), *Fagus sylvatica*, *Populus tremuloides*, *Pinus ponderosa* and *Picea rubens* (cited in Körner 2012).

While both SLA and chlorophyll concentrations are usually used as performance indicators (e.g., Cornelissen *et al.* 2003), in our study neither trait matched with variations in sapling performance. The direction of SLA variation recorded for *P. australis* saplings in our study was unexpected in that the unimodal pattern of sapling performance was not coupled with a common SLA trend but with an opposite pattern, with the lowest SLA values being recorded for saplings from intermediate altitudes (i.e., those with highest sapling and leaf biomass, LA and height). This SLA trend could be related to the unimodal reduction in LA toward the altitudinal limits of the species distribution. A reduction in LA without an increase in leaf thickness (not measured here) results in a reduction in leaf mass, which in turn can result in a surprisingly high SLA (Witkowski & Lamont 1991; Wilson, Thompson & Hodgson 1999). Another explanation for the observed pattern is that high SLA values under stressful conditions may facilitate rapid growth responses under short favorable periods (Gotsch, Powers & Lerdau 2010), such as wet periods at lower altitudes or warm temperatures at higher altitudes. However, no such rapid response was recorded in our study under the favorable greenhouse conditions during the 18 months of our experiment. Although SLA is a commonly used trait to explain interspecific differences in seedling performance (Cornelissen *et al.* 2003), our results show the need to be cautious at drawing conclusions at least in this mountain system, since variation in SLA among populations of *P. australis* would not properly reflect variations in sapling performance, as discussed by Sánchez-Gómez, Zavala & Valladares (2008) and Stahl *et al.* (2014).

Sapling biomass is usually related to chlorophyll concentration because the latter trait is an integrative measure of overall leaf functional status associated with leaf structure, N and pigment concentrations (Kitajima & Hogan 2003; Díaz-Espejo, Nicolás & Fernandez 2007; Martin, Asner & Sack 2007). However, in our study, while *P. australis* saplings from intermediate and higher altitudes showed no differences in chlorophyll concentration, they did show different height and biomass levels. Similarly, sapling biomass in *M. boaria* and *E. cordobensis* populations varied with altitude, with no differences being observed in chlorophyll concentration. At higher altitudes, an increased allocation to underground structures (Hertel & Wesche 2008) or a trade-off between relative growth rate and frost resistance (Körner 2012) may explain the lack of relationship between both traits. Additionally, variation in leaf density and leaf thickness can affect photosynthetic rates (Sack *et al.* 2013). Dense leaves can reduce photosynthetic rates by decreasing assimilative leaf compounds and limiting the diffusion of CO₂ and penetration of light into the leaf, thereby modifying the relationship between photosynthesis and SLA (Niinemets 1999).

Remarkably, seed mass might affect early seedling development (Fenner & Thompson 2005). The initial differences in seedlings among seeds of different size could achieve parity after different periods, such as 105 or 160 days after germination (Fenner & Thompson 2005). In this study, the 18-month period under greenhouse conditions seemed to be long enough to overcome this potential effect of seed mass at least for *M. boaria* and *E. cordobensis*. However, this effect cannot be discarded for *P. australis*, since seed mass and some sapling traits followed a similar altitudinal pattern. Future studies should include the influence of seed mass on sapling traits.

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Finally, genetic differences between traits described for many woody species occurring at high altitudes are understood to be a result of directional selection which may promote populations from such high altitudinal limits to exhibit lower phenotypic plasticity in growth and morphological traits in response to temperature changes than populations growing at low altitudes (Debat & David 2001; Vitassé *et al.* 2013). In our study, saplings of the three species from both altitudinal extremes tended to exhibit the overall lowest performance (i.e., at least in terms of biomass). Indeed, saplings from lower and normally dryer altitudes, and those from upper and normally cooler altitudes were not able to take advantage of the wetter and milder environment of the greenhouse. Further studies should be conducted to determine whether contrasting environmental stress factors at the altitudinal distribution extremes in such systems might reduce the phenotypic plasticity of these woody species, which need to be assessed including intrapopulation variability.

Conclusion

Populations in rapidly changing environments can persist by either adapting to conditions in their usual habitat or migrating to other areas; otherwise, they may become locally extinct (Aitken *et al.*, 2008; Nicotra *et al.*, 2010). Our findings in relation to the very early phases of the plant life cycle suggest that both germination success and sapling performance of the main woody species of the Sierras Grandes Mountains appear to be well adjusted to their current maternal environments, which may be detrimental under climate warming scenarios (Herman & Sultan 2011), especially for *P. australis*, which seems to have a broader number of adjusted sapling traits. Additional studies are necessary to explore the influence of differences among individuals in the patterns observed and the adaptive value of the observed

intraspecific variability in our study species, which could be evaluated through reciprocal transplants and specific genetic studies (González-Martínez, Krutovsky & Neale 2006).

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Figure 1. Seed germination success (% Mean $\pm$ SE) in response to three different temperature treatments for (a) *Polylepis australis*, (b) *Maytenus boaria* and (c) *Escallonia cordobensis*. Letters indicate mean differences between altitudes*temperature regimes (DGC, $p < 0.05$).

Figure 2. Attribute variation in seeds from different altitudes and saplings grown in the greenhouse for 18 months measured for *Polylepis australis*. Means $\pm$ SE are indicated. Number of replications (N) per elevation is indicated in each figure.

Figure 3. Attribute variation in seeds from different altitudes and saplings grown in the greenhouse for 18 months measured for *Maytenus boaria*. Means $\pm$ SE are indicated. Number of replications (N) per elevation is indicated in each figure.

Figure 4. Attribute variation in seeds from different altitudes and saplings grown in the greenhouse for 18 months measured for *Escallonia cordobensis*. Means $\pm$ SE are indicated. Seed mass of *E. cordobensis* shows three values per elevation; each value corresponds to 200 seeds (see methods for details). Number of replications (N) per elevation is indicated in each figure.

Table 1. Variation in seed and sapling traits of the three study species among populations from different altitudes on 18-month-old saplings cultivated in greenhouse. Test value shows F values corresponding to ANOVA using General Linear Models; analyzed deviance values corresponding to Generalized Linear Models for mortality are shown. Degree of freedom of General and Generalized Linear Models was 6 for *Polylepis australis* and 3 for *Maytenus boaria* and *Escallonia cordobensis*. The other columns summarize the explained proportion of variance (adjusted R^2) in intra-specific variation of functional traits in linear regressions or binary logistic regression and slope for altitude and their squared term (-, negative; +, positive; $\cap$, unimodal; U, u pattern). For mortality, Nagelkerke R^2 is reported. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns, $p > 0.05$.

Trait	<i>Polylepis australis</i>				<i>Maytenus boaria</i>				<i>Escallonia cordobensis</i>			
	Linear Model	Regression			Linear Model	Regression			Linear Model	Regression		
Seed trait	Test value	R^2	Altitude		Test value	R^2	Altitude		Test value	R^2	Altitude	
Seed mass	15.24***	0.31	+	***	49.81***	0.56	-	***	6.27*	0.70	-	*
Sapling traits												
Plant biomass	6.11***	0.062	+	*	4.57**	0.045	ns	ns	2.06 ns	0.014	+	*
Height	8.08***	0.115	+	**	6.23***	0.03	+	**	0.14 ns	0	ns	ns
Leaf area	4.38***	0.044	+	*	4.46**	0.052	ns	ns	2.07 ns	0	ns	ns
SLA	3.60**	0.047	-	**	0.60 ns	0	ns	ns	0.85 ns	0.001	ns	ns
Leaf biomass	12.09***	0.118	+	***	4.44**	0.051	ns	ns	2.56 ns	0.002	ns	ns
Chlorophyll	2.60*	0.021	+	**	1.01 ns	0.004	ns	ns	2.27 ns	0.024	ns	ns
Mortality	27.31***	0.082	-	*	1.31 ns	0.01	ns	ns	3.76 ns	0.31	ns	ns

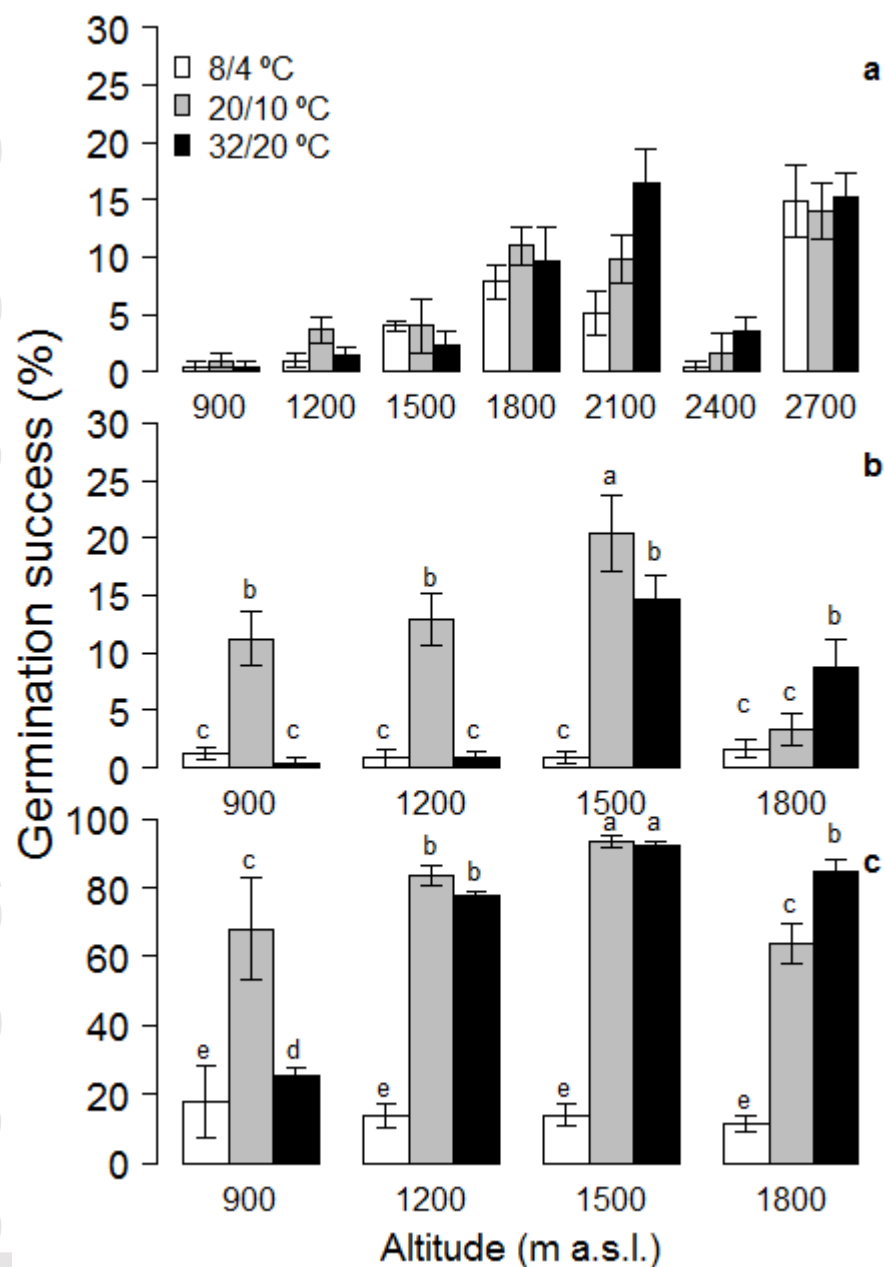


Fig. 1

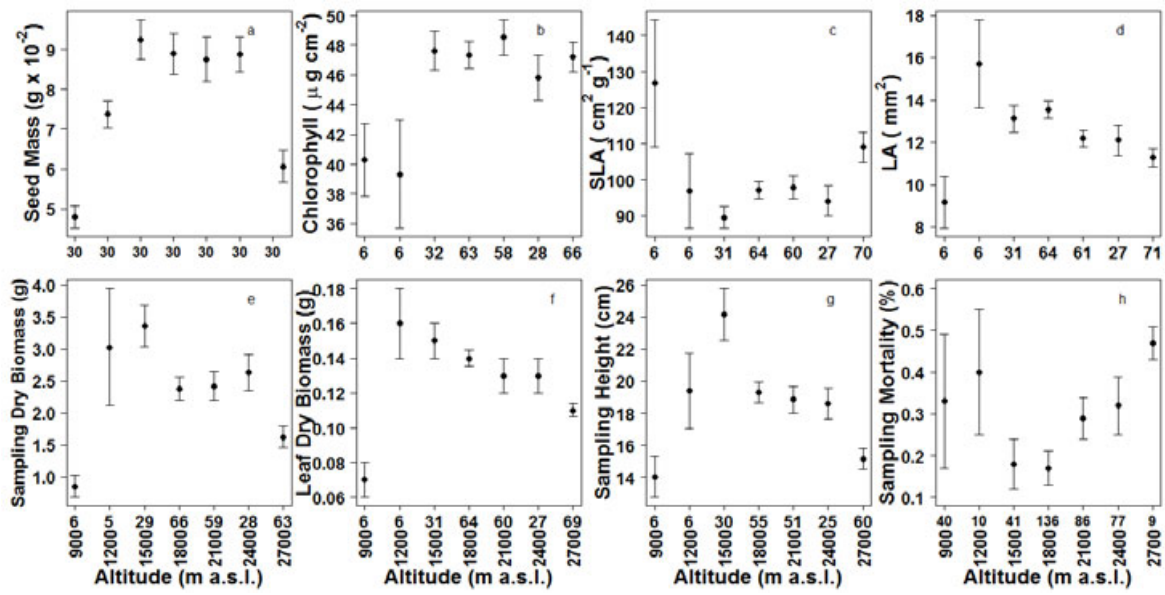


Fig. 2. *Polylepis australis*

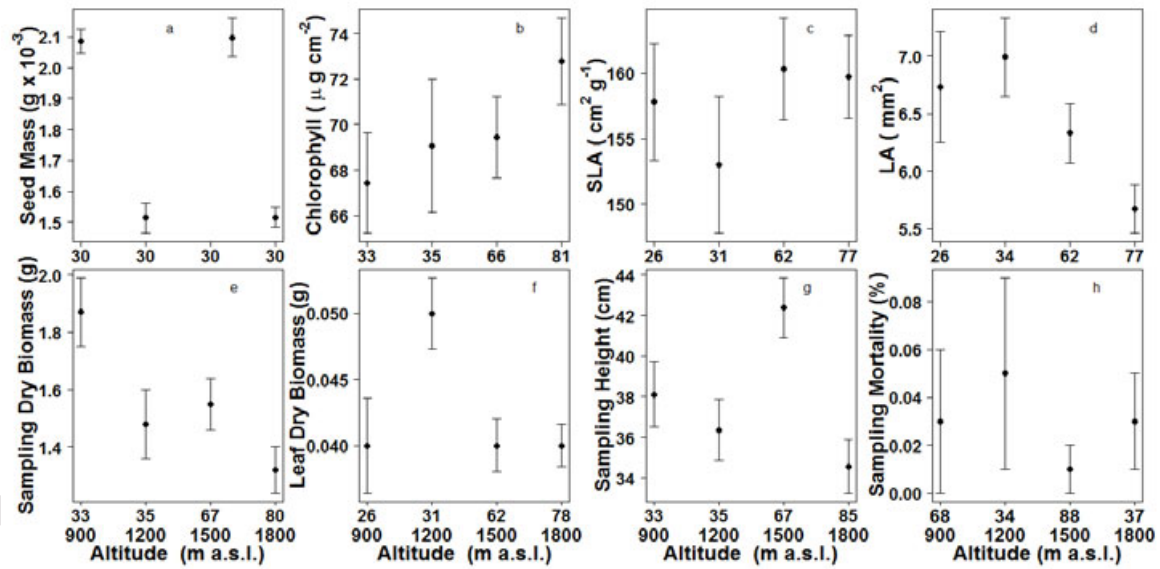


Fig. 3. *Maytenus boaria*

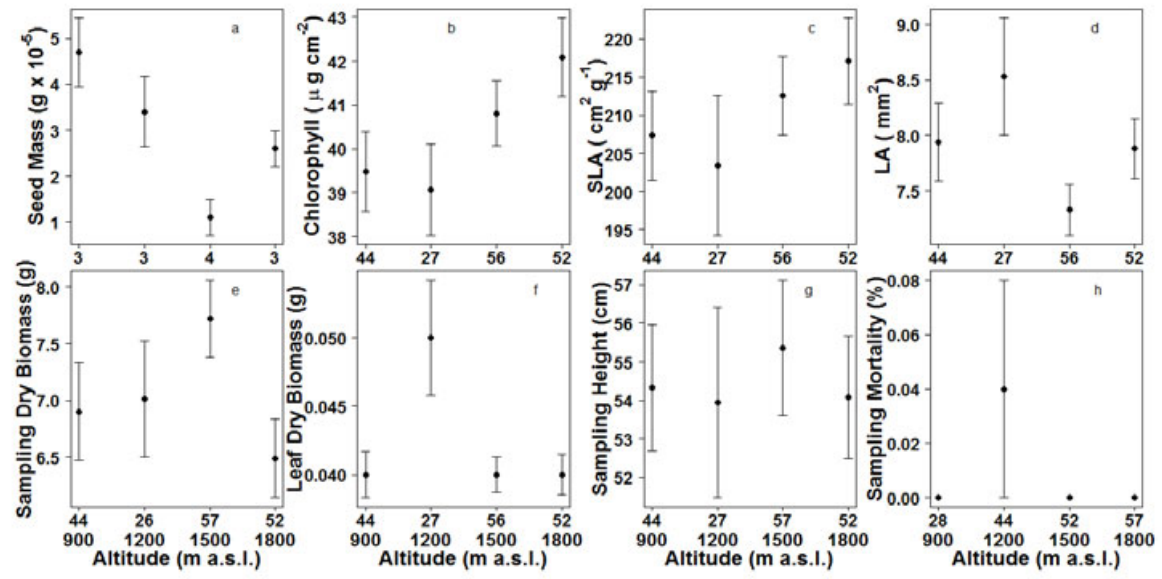


Fig. 4. *Escallonia cordobensis*